Multi-Omics Target Discovery Package

Lung Adenocarcinoma (LUAD) *BIOT 6900 · Portfolio Deliverable 1*

Bhuvanesh Karthik Nilavalagan

1. Disease and Question

Lung adenocarcinoma is the most frequent type of lung cancer and accounts for a considerable number of cancer-related deaths across the globe. Unlike other cancers, lung adenocarcinoma has a well-studied data environment like the same patients have been characterized both at the level of RNA expression, protein abundance, and genome alterations, which allows asking whether a proposed drug target is confirmed by independent types of biological data. A drug target in such a setting is a gene that is repeatedly found to be disrupted on multiple biological levels, looks like a class of proteins known to be amenable to therapeutic intervention, and has a true binding site available once it has been structured. This approach takes you through all three filters in one example of lung adenocarcinoma target discovery.

2. Multi-Omics Integration

Three independent data layers were combined for this disease: transcriptomic and proteomic measurements from 213 matched tumour and normal tissue samples (111 tumour, 102 matched-normal), and genomic risk variants from a genome-wide association study of lung adenocarcinoma. Each gene's transcriptomic and proteomic change was tested for statistical significance between tumour and normal tissue, and every gene with a reported genetic association was retained as genomic evidence. Harmonizing all three layers onto a common gene symbol, after resolving inconsistent identifier formats across the three source databases, left 154 genes with usable evidence in all three layers at once; this is the pool the final ranking was built from.

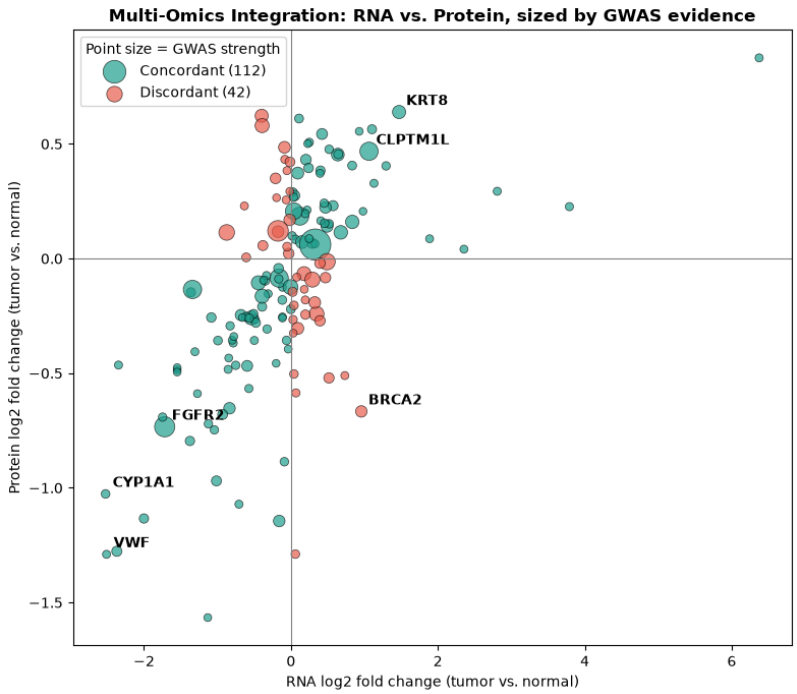


Figure 1. Multi-omics integration across 154 harmonized genes. Each point is one gene, plotted by its tumour-versus-normal change at the RNA level (x-axis) and protein level (y-axis), with point size reflecting the strength of its genetic association with lung adenocarcinoma.

Of those 154 genes, 112 (73%) showed RNA and protein changing in the same direction, a pattern called concordance. The remaining 27% moved in opposite directions, which is not noise: it is the signature of post-transcriptional regulation, where a cell changes how quickly it builds or destroys a protein independently of how much RNA it is making. Reading this disagreement rather than discarding it is discussed further in Section 6.

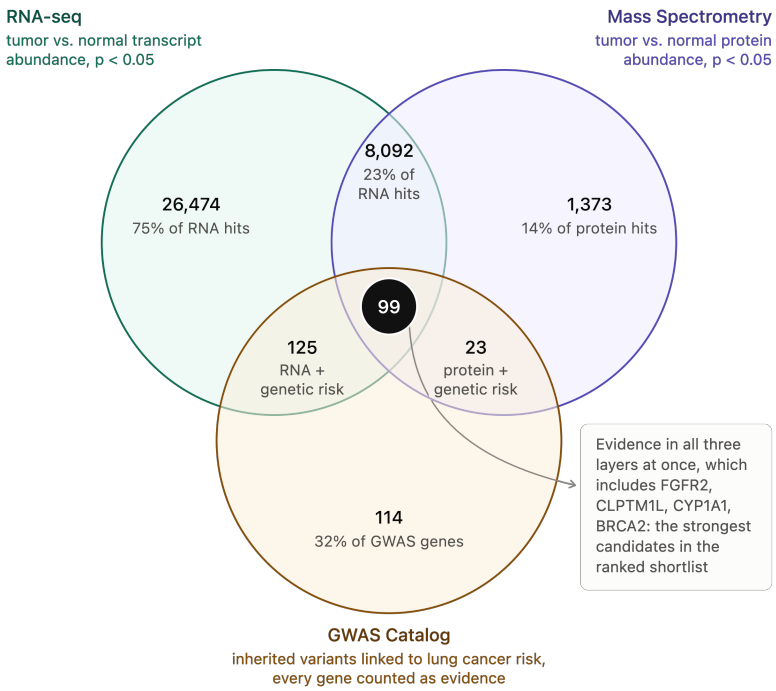


Figure 2. Venn Diagram- Cross-layer evidence overlap. 99 genes cleared an independent significance threshold in all three layers simultaneously, the strongest form of convergent evidence this pipeline can produce.

Each gene's evidence across the three layers was combined into a single multi-evidence score, giving transcriptomic, proteomic, and genomic evidence equal weight. This avoids letting one layer's scale dominate simply because its numbers happen to be larger.

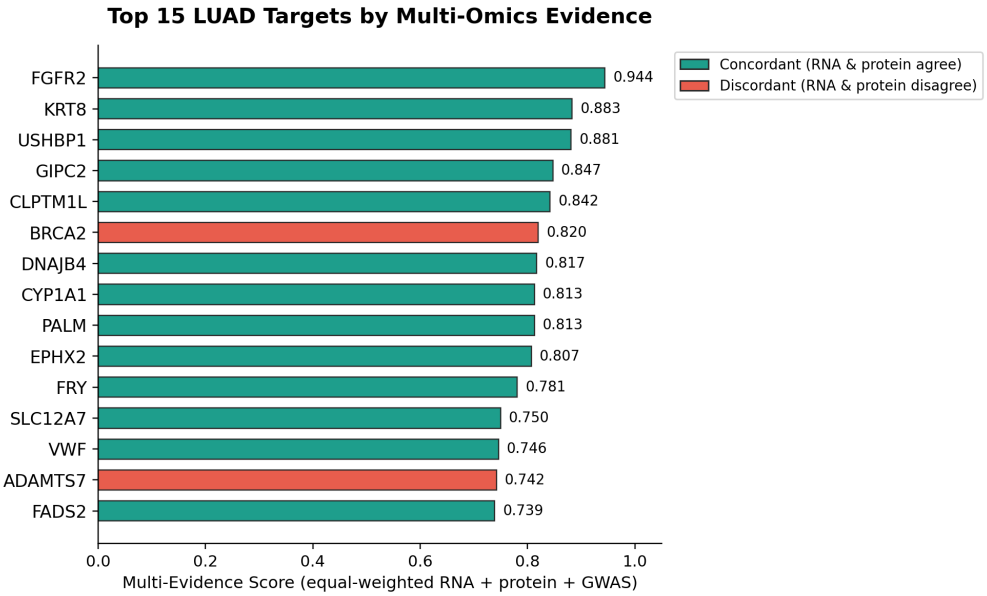


Figure 3. Top 15 ranked LUAD targets. FGFR2 leads with strong, concordant evidence across all three layers; BRCA2 and ADAMTS7 (red) are the two discordant genes in the top 15.

The discordant genes (42) where RNA and protein disagree deserve a note before moving on. The most clear example is provided by BRCA2 because its RNA levels increase in tumors but its protein decreases. Because BRCA2 repairs DNA damage and the amount of it is strictly regulated, the increased expression level does not lead to an increase in active protein. Such genes were left because their contradictory expression levels show that some kind of regulation takes place after the transcription. There is one other feature which is important for all following considerations. The score depends on the amount of change in one or another way, which results in some of the top genes being significantly decreased in tumors.

3. Druggability Triage

A gene being disrupted in disease does not automatically mean a drug can be made against it. To check this, each of the top 15 targets was compared against a large reference panel of proteins with known clinical drug history, using ESM C protein language model to ask whether each target's sequence resembles a protein class that has already been successfully drugged, and whether that resemblance points toward a small molecule or an antibody.

This comparison was validated honestly rather than optimistically: a random split of the reference panel made the model look very accurate, but that number is inflated because close relatives of a held-out protein often remain in the training data. Holding out entire protein families at once, a harder and more realistic test, gave a lower but more trustworthy accuracy. That family-aware number is the one relied on for every tractability claim in this package.

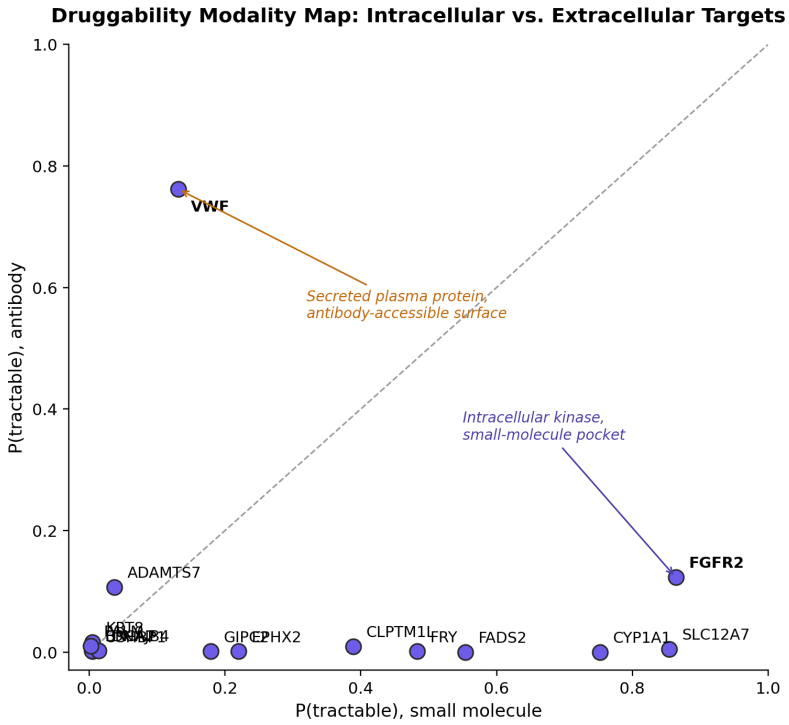


Figure 4. Druggability modality map. FGFR2, an intracellular kinase with an enzymatic pocket, scores strongly for small-molecule drugs, while VWF, a large secreted blood protein, scores strongly for antibodies instead. Modality tracks where a protein sits and what shape it takes, not just how strongly it is linked to disease.

The map shows that tractability depends on where a protein lives and what shape it takes, not just how strongly it is linked to disease. FGFR2, an intracellular kinase whose closest drugged relative is the near-identical receptor FGFR1, sits firmly on the small-molecule side, as does CYP1A1, which most resembles the established drug target CYP17A1. VWF, a large secreted blood protein, points instead toward an antibody approach. Several genes with strong disease evidence, including BRCA2, KRT8, and PALM, cluster near zero on both axes, meaning nothing in their sequence resembles a protein that has ever been successfully drugged. Combining this prior with the original evidence, weighted 60 to 40 in Favor of the patient data, kept FGFR2 at the top while BRCA2 dropped from sixth to twelfth.

It is worth being clear about what this stage settles. The model learns from the history of drug discovery, not from physics, so a protein scoring near zero is not necessarily undruggable; it may simply belong to a class nobody has cracked yet. BRCA2's low score reflects the long-standing difficulty of drugging DNA-repair scaffolds rather than any measured property of the protein. The prior was therefore weighted below the patient data and treated as a filter, not a verdict. The two highest scorers were carried into structure prediction precisely because sequence resemblance, however suggestive, says nothing about whether a real pocket physically exists.

4. Structural Assessment

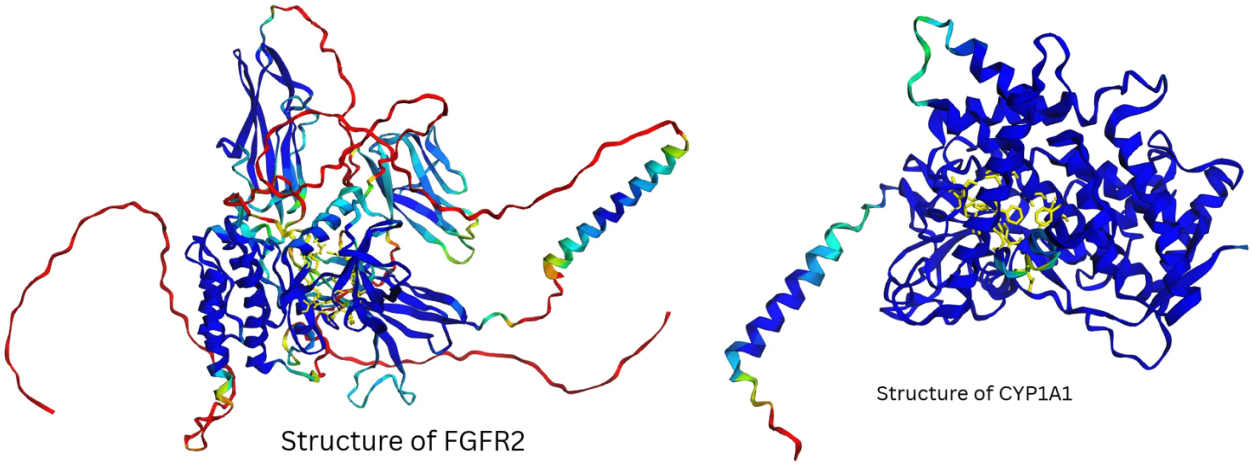


Figure 5. AlphaFold structures of FGFR2 and CYP1A1 (py3Dmol), coloured by confidence (blue = high, red = low)

The two strongest small-molecule candidates, FGFR2 and CYP1A1, were carried forward to structure prediction using Alphafold. Each protein's predicted 3D structure was generated and checked two ways: how confident the prediction is at each position (a measure called pLDDT), and how well the predicted structure matches a real, experimentally solved structure of the same protein.

Both predictions held up well against real laboratory data. CYP1A1 matched its solved crystal structure closely across nearly its entire length. FGFR2's kinase domain, the only piece with a matching experimental structure available, also matched closely; the rest of the receptor has no solved experimental structure to compare against, a known limitation of structural biology for large multi-domain receptors generally, not a flaw specific to this analysis.

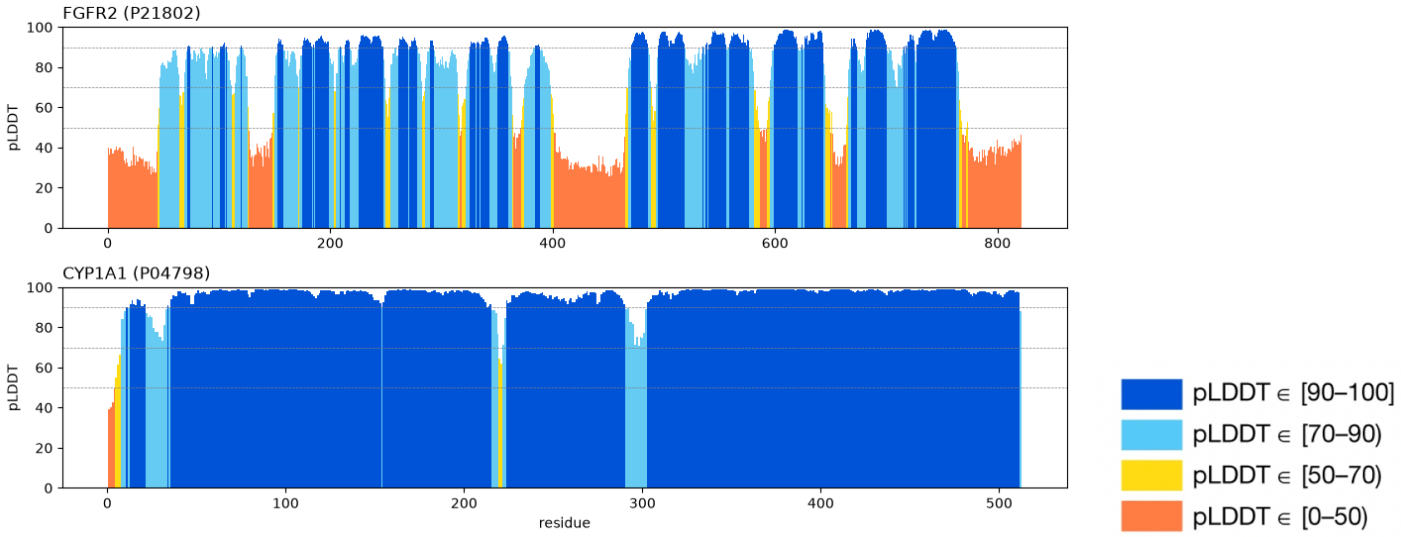


Figure 6. Per-residue confidence (pLDDT) for FGFR2 (top) and CYP1A1 (bottom). CYP1A1 is a single, uniformly confident enzyme domain. FGFR2 shows confident domains separated by genuinely flexible linker regions, consistent with it being a multi-domain membrane receptor rather than one rigid block.

Searching each confident structure for a real drug-binding pocket found one strong, trustworthy candidate pocket in each protein, and in both cases that pocket was independently confirmed to be exactly where a real, already-known drug molecule physically sits in the experimental structure. A second candidate pocket in FGFR2 that looked promising by shape alone was rejected once confidence was checked, since it sat in one of the protein's flexible, low-confidence regions, a concrete demonstration that geometry alone can be misleading without a confidence check behind it.

5. Prioritized Target Package

Combining all three layers of evidence into one shortlist, FGFR2 is the clearest candidate: it led the original multi-omics ranking, showed the strongest small-molecule resemblance in the tractability stage, and is now the only target with a confirmed, drug-occupied structural pocket. CYP1A1 follows a similar path with a slightly lower starting rank but an equally strong tractability and structural result, and connects mechanistically to tobacco carcinogen metabolism, a plausible route from environmental exposure to lung cancer risk. VWF represents a genuinely different kind of opportunity: strong multi-omics evidence paired with an antibody-favouring structure rather than a small-molecule one. BRCA2, despite strong multi-omics evidence, showed essentially no resemblance to any known drug target and would need a different strategy entirely, such as a pathway-level or synthetic-lethality approach, rather than a direct small-molecule or antibody effort.

Rank	Gene	Multi-omics score	Tractability	Structure checked	Final call for Discovery team
1	FGFR2	0.944	0.864 (small molecule)	Yes - confirmed pocket	Small-molecule lead
2	KRT8	0.883	0.016 (weak, either)	Not yet assessed	Carry forward, re-screen
3	CLPTM1L	0.842	0.389 (moderate, SM)	Not yet assessed	Carry forward
4	CYP1A1	0.813	0.752 (small molecule)	Yes - confirmed pocket	Small-molecule lead
5	VWF	0.746	0.762 (antibody)	Not yet assessed	Antibody candidate
6	BRCA2	0.820	0.004 (negligible)	Not yet assessed	Deprioritize for SM/Ab; consider pathway strategy

6. Limitations and Validation

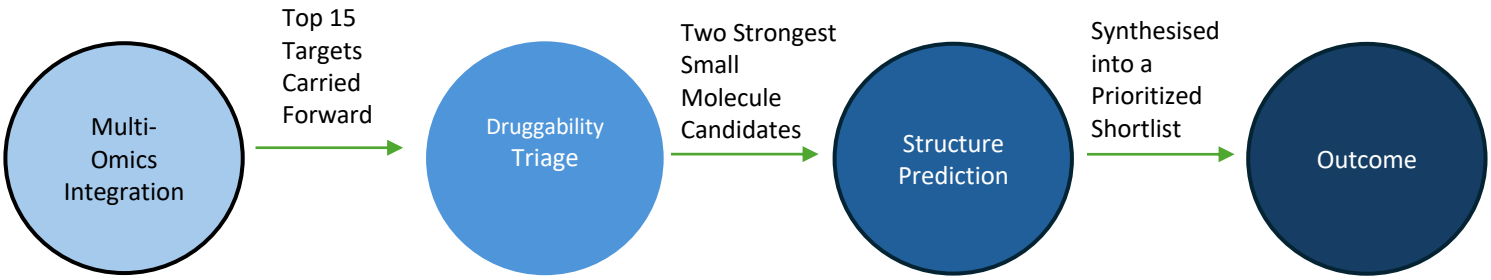
In this study, a set of genes having convergent statistics and structure is detected. It cannot be assumed that any of the detected genes causes lung adenocarcinoma or that there will be an effective medication for treatment through the use of these genes. The genomic level of data collection was from a completely different pool of subjects to the 213 patients from whom RNA and protein data was collected.

Smoking status is invisible in this analysis. Lung adenocarcinoma in smokers and never-smokers are close to different diseases at the molecular level, with different driver mutations and different biology. Averaging across all 213 patients could easily mask a gene that matters intensely in one group and not at all in the other, and it may be part of why several top candidates appear decreased on average.

The conflicting signal of BRCA2, where there is an increase in RNA but a decrease in the protein, is viewed here as a signal rather than noise because BRCA2 is a DNA repair protein, and it is common knowledge that cells strictly regulate the levels of DNA repair proteins after synthesis, thus accounting for the lack of increase in protein levels despite the increased RNA levels. Predicted protein structure is just one snapshot of the structure and cannot determine if the pocket will remain open after binding to the actual partners, or if the cancer-causing mutation can form a pocket not present in the structure.

The next experiment that would most directly validate the top candidate is a functional assay on FGFR2: expressing the protein, confirming the predicted pocket binds a known FGFR inhibitor in solution, and testing whether inhibiting it reduces proliferation in a LUAD cell line carrying the same expression pattern seen in this analysis. A positive result would move FGFR2 from a computationally convergent candidate to an experimentally validated one, the next real step for a discovery team receiving this package.

7. Target Pipeline



Reference:

1. McKay, J. D., Hung, R. J., Gaborieau, V., Boffetta, P., Chabrier, A., Byrnes, G., Zaridze, D., Mukeria, A., Szeszenia-Dabrowska, N., Lissowska, J., Rudnai, P., Fabianova, E., Mates, D., Bencko, V., Foretova, L., Janout, V., McLaughlin, J., Shepherd, F., Montpetit, A., … Brennan, P. (2008). Lung cancer susceptibility locus at 5p15.33. https://doi.org/10.1038/ng.254

   (Used to confirm that the 5p15.33 locus, which contains my fifth-ranked gene CLPTM1L, is a genuine and independently replicated lung cancer susceptibility region)

2. James, M. A., Wen, W., Wang, Y., Byers, L. A., Heymach, J. V., Coombes, K. R., Girard, L., Minna, J., & You, M. (2012). Functional characterization of CLPTM1L as a lung cancer risk candidate gene in the 5p15.33 locus. https://doi.org/10.1371/journal.pone.0036116

3. Hukkanen, J., Pelkonen, O., Hakkola, J., & Raunio, H. (2002). Expression and regulation of xenobiotic-metabolizing cytochrome P450 (CYP) enzymes in human lung. https://doi.org/10.1080/20024091064273

   (Used to support the claim that CYP1A1 metabolically activates tobacco carcinogens in human lung tissue

4. Pacini, L., Jenks, A. D., Lima, N. C., & Huang, P. H. (2021). Targeting the fibroblast growth factor receptor (FGFR) family in lung cancer. https://doi.org/10.3390/cells10051154

   (Used to confirm that FGFR family receptors, including my top-ranked target FGFR2, are established small-molecule drug targets in lung cancer)

5. Gillette, M. A., Satpathy, S., Cao, S., Dhanasekaran, S. M., Vasaikar, S. V., Krug, K., Petralia, F., Li, Y., Liang, W.-W., Reva, B., Krek, A., Ji, J., Song, X., Liu, W., Hong, R., Yao, L., Blumenberg, L., Savage, S. R., Wendl, M. C., … Clinical Proteomic Tumor Analysis Consortium. (2020). Proteogenomic characterization reveals therapeutic vulnerabilities in lung adenocarcinoma. https://doi.org/10.1016/j.cell.2020.06.013